

Submission Summary

Conference Name

International Conference on Next Generation Electronics

Paper ID

3623

Paper Title

Evidence-Guided Multimodal Fake News Detection using Text-Image Consistency and External Knowledge

Abstract

Although Social media platforms are now of the main channels for exchanging information, they have also hastened the spread of false information and deceptive content. Because contemporary misinformation frequently blends textual assertions with altered or deceptive visual content, it is difficult to identify fake news in such settings. As a result, traditional text-only detection techniques frequently lack trustworthy procedures for verifying claims against outside evidence and struggle to identify multimodal contradictions. In order to detect fake news, this paper suggest an Evidence-based multimodal approach that incorporates external knowledge verification, image-text consistency analysis and textual semantics. To assess the semantic congruence between news claims and its related photos, the system first uses contextual embeddings created by Sentence-BERT to represent textual information. Visual features then processed by a CLIP-based encoder. A retrieved-based verification module is developed to further evaluated the information's credibility. Using FAISS Vector search and SBERT embeddings, pertinent evidence is retrieved from an external knowledge source. A Natural language inference(NLI) Model based on BART-Large-MNLI is then used to analyze the recovered evidence in order to ascertain whether the claim is supported, refuted, or unrelated to the material at hand. A Feature fusion process is used to merge the features obtained from text semantics, Multimodal consistency, and Evidence verification and LightGBM model is used for classification. The suggested framework performs well across common classification criteria. The reliability of detecting fake news is further enhanced by combining multimodal signals with retrieved-based evidence verification.

Keywords - Fake news detection, Multimodal Learning, Retrieval-Augmented Generation(RAG), Text-Image consistency, Natural Language Inference, Evidence verification, LightGBM, Social media analysis

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Primary Subject Area

Special Session 2 - Bridging the Digital Divide in the Age of Artificial Intelligence: Leveraging Explainable AI for Social Good

Submission Files

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